Rates and Unit Rates Flagship

Lesson 4-1-flagship

Name: ____________ Date: ____________ Class: ____________

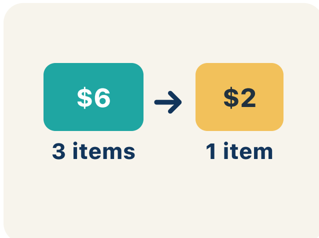
ARCADE BUILDER MISSION

Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players — Booth A charges $3 for 5 games, Booth B charges $5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

Key Vocabulary (Level 1 support)

Picture first, then the word, then a plain-language meaning. Say each word out loud.



$12 for 4 games is a rate

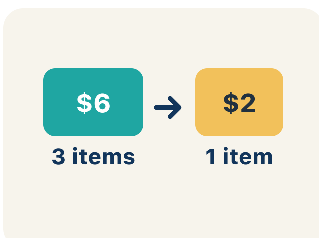
Rate

A ratio comparing two amounts with different units, like miles per hour.

$3 per 1 game

Unit Rate

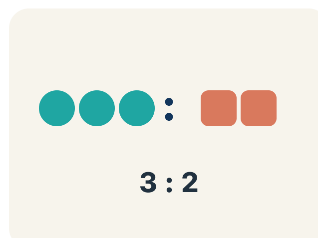
A rate for just 1 of something, like cost for 1 item.



60 miles per hour means 60 miles for every 1 hour

Per

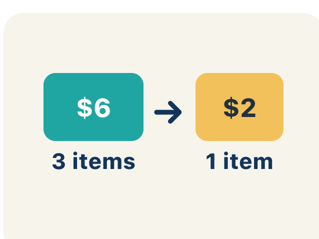
For each one. Example: 5 dollars per book.



5 to 3 or 5:3

Ratio

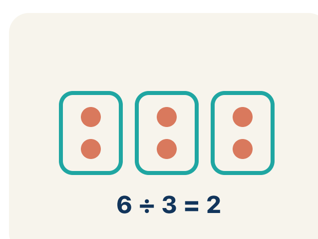
A way to compare two amounts.



Pack A: $0.50 per pencil vs Pack B: $0.40 per pencil — Pack B is the better buy

Better Buy

The choice that costs less for each item.



$12 ÷ 4 games = $3 per game

Divide

Splitting a total into equal groups.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find a unit rate to compare prices and decide the better buy.

1️⃣ Notice

You're comparing ticket prices at different game booths in the arcade.

2️⃣ Key idea

I can find a unit rate to compare prices and decide the better buy.

3️⃣ Words I'll use

Rate **Unit Rate** **Per** **Ratio** **Better Buy** **Divide**

🤔 Think About It

- What two quantities are being compared at each booth?
- Which booth seems like a better deal at first glance?
- How can you compare prices when the number of games is different?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: A booth charges $4.50 for 9 games. What is the unit rate?

A. $0.50 per game

B. $0.45 per game

C. $2.00 per game

D. $4.50 per game

Step 1 $4.50 ÷ 9 = $0.50 per game.

✅ **Answer:** A. $0.50 per game

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

A. 30 miles per gallon

B. 6 miles per gallon

C. 180 miles per gallon

D. 36 miles per gallon

Step 1 ______

Step 2 ______

Answer: ______

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: A pack of 8 markers costs $6.00. What is the cost per marker?

A. $0.75

B. $0.80

C. $1.33

D. $0.60

Show your work:

______

______

______

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Booth A charges $3 for 5 games and Booth B charges $5 for 8 games. Can you tell which is the better deal just by looking at the signs? Why or why not?

Level 1 support

Start here: You can't compare fairly until you find the cost per 1 game (the unit rate) for each booth.

💡 **Need a hint?**

Sentence starters (optional)

You ____ tell just by looking because the number of ____ is different.

____ se puede saber solo mirando porque la cantidad de ____ es diferente.

To compare fairly I need the cost per ____ game.

Para comparar de forma justa necesito el costo por ____ juego.

Word bank: rate | unit rate | per | better buy | compare

Listen for: *Students recognize the booths offer different numbers of games, so a fair comparison needs the cost per 1 game (unit rate).*

Level 2

Some players pick Booth B because $5 buys more games. Why is that reasoning not enough to know the better deal? Justify using cost per game.

- Buying more games does not always mean a better deal because ____.
- The real test is the ____, since ____.

EXPLORE

How did you find the cost per game for each booth, and which booth wins?

Level 1 support

Start here: Divide the price by the number of games to get the unit rate, then compare the two rates.

💡 **Need a hint?**

Sentence starters (optional)

Booth A costs ____ per game because 3 divided by 5 equals ____.

El puesto A cuesta ____ por juego porque 3 entre 5 es ____.

The better buy is Booth ____ because its ____ is lower.

La mejor compra es el puesto ____ porque su ____ es más baja.

Word bank: rate | unit rate | per | better buy | divide

Listen for: *Students compute $0.60/game for Booth A and $0.625/game for Booth B, then choose Booth A because the lower unit rate is the better buy.*

Level 2

If a player only wants 5 games total, does the better unit rate always give the cheaper total? Explain when it might not.

- The lower unit rate gives the cheaper total only when ____.
- It might not be cheaper if ____.

CONNECT

At the snack bar, how would you use a unit rate to find the best popcorn deal among different sizes?

Level 1 support

Start here: Find the price per ounce (or per cup) for each size, then pick the lowest unit rate.

💡 **Need a hint?**

Sentence starters (optional)

I would find the price per ___ for each popcorn size.

Hallaría el precio por ___ para cada tamaño de palomitas.

The best deal is the one with the ___ unit rate.

La mejor oferta es la que tiene la ___ tasa unitaria.

Word bank: rate | unit rate | per | better buy | compare

Listen for: *Students describe dividing price by size to get a per-unit cost and choosing the lowest unit rate as the best deal.*

Level 2

Why might the biggest popcorn not always be the best value? Use the idea of unit rate to explain.

- The biggest size is not always the best value because ___.
- The unit rate shows the real value because ___.

Write About the Math

The Writing Revolution

I can explain my choice using the words rate, unit rate, per, and better buy.

1. Kernel Sentence

subject + verb

Model: Rate is a ratio comparing two amounts with different units, like miles per hour.
Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.

Write a kernel sentence about rate. Use a subject and a verb.
Escribe una oración base sobre tasa. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Rate matters in math
Tasa importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Rate matters in math because ____.** *Tasa importa en matemáticas porque ____.* ___
but *pero*	**Rate matters in math, but ____.** *Tasa importa en matemáticas, pero ____.* ___
so *entonces*	**Rate matters in math, so ____.** *Tasa importa en matemáticas, entonces ____.* ___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about rate.
Di un hecho verdadero sobre rate.

Rate ____.

Question
Pregunta

Ask a question about rate.
Haz una pregunta sobre rate.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about rate.
Muestra entusiasmo sobre rate.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with rate.
Dile a un compañero qué hacer con rate.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

You ____ tell just by looking because the number of ____ is different.
____ se puede saber solo mirando porque la cantidad de ____ es diferente.

To compare fairly I need the cost per ____ game.
Para comparar de forma justa necesito el costo por ____ juego.

Booth A costs ____ per game because 3 divided by 5 equals ____.
El puesto A cuesta ____ por juego porque 3 entre 5 es ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which token pack from the table above is the best deal?

A. Pack A

B. Pack B

C. Pack C

D. They are all the same

Show your work:

2. Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

A. Runner B

B. Runner A

C. Same rate

D. Cannot determine

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for $5, 60 tokens for $10.80, and 100 tokens for $19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of $____ per token.

Show your work:

Reflect — Exit Ticket

A store sells 6 pencils for $1.50 and 10 pencils for $2.80. Which is the better buy?

A. 6 for $1.50 — $0.25 each

B. 10 for $2.80 — $0.28 each

C. They cost the same per pencil

D. Not enough information

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 30 miles per gallon *— 180 miles ÷ 6 gallons = 30 miles per gallon.*
2. **You Try (notes):** A. $0.75 *— $6.00 ÷ 8 = $0.75 per marker.*
3. **Try It 1:** B. Pack B *— Pack B has a unit rate of $0.20 per token, which is the lowest cost per token.*
4. **Try It 2:** A. Runner B *— Runner A: 4÷12 ≈ 0.33 laps/min. Runner B: 5÷14 ≈ 0.36 laps/min. Runner B runs slightly more laps per minute.*
5. **Exit Ticket:** A. 6 for $1.50 — $0.25 each *— $1.50 ÷ 6 = $0.25 per pencil. $2.80 ÷ 10 = $0.28 per pencil. $0.25 < $0.28, so 6 for $1.50 is the better buy.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).